Title: Advanced Microcontroller Based Fault Protection Device for Home Appliance.

Introduction: In residential area fault from power grids such as over voltage and overcurrent can result in permanent damage of home appliance it also can be result in life threatening accident. This project proposes an automated, low-cost user-friendly protection system that can detect overvoltage, overcurrent and take steps to safeguard appliance and can do restart when normal condition is achieved without any humane interaction ensuring 24X7 monitoring. The proposed system is capable of detecting abnormal or sudden rise in voltage or current and can perform a shutdown to safeguard the appliance and once the normal condition arrives it can restart the appliance in operating state again. For repetitive fault detection it also has a lockdown mechanism that will protect the device from permanent damage. All status about power condition will be indicated or notified to the used via 16x2 LCD display and LEDs as indicator. The advance feature includes user control, where a user can modify threshold value by using buttons and LCD display.

Components: Arduino MEGA 2560, ACS712-30A sensor, LED, LCD display, Push-Button, Resistor, Capacitor.

System Architecture: Arduino Mega 2560 is the microcontroller that work as a brain in this system. The other units of the project are voltage sensing, load current sensing and user-interface. The Voltage sensing unit and current sensing unit feed analog signal to the microcontroller ADC pins A0 and A1 respectively. ADC pins convert analog inputs to digital values and do the mathematics calculation at the software side to measure input voltage and load current. At the software end these values are used to evaluate set logic and determine the power condition and according the logic microcontroller perform specific task to ensure proper system operation.

Implementation and Protection Logic: Since this project involved AC power without expert guidance or observation simulation for now was the safest approach to take. The project was simulated on advance simulation software proteus 8. To simulate embedded project in proteus external library for components are required to be downloaded. For this project I have worked on building the voltage sensing unit from scratch rather downloading any existing library since I didn't find a reliable library that can simulate accurate AC voltage input. For voltage sensing a circuit was developed consisting coupling transformer that can lower 220V AC to 9V ac a voltage divider that scale voltage approx. 0.1279 ratio and 2.5V bias network to make proper input for ADC that ranges from 0 to 5V. For current sensing ACS712-30A sensor library was used. For calculating both RMS voltage and RMS current via software code involved mathematical calculation one important value that was calibrated for voltage sensing was (5/1024) ÷ 0.1279 (divider ratio) × 24.44 (transformer ratio)=0.933 but in real simulation practice for minimum error have to modify the value to 0.9421 and for current sensor used the divider ratio 0.066 that was referred on internet and (5/1024) ÷ 0.066=0.0739 here used the exact value on library code. The logic was if fault occurs microcontroller send a signal to mechanical relay via it's pin and load get disconnected and when voltage and current enter safe region it takes a delay (in code 5 sec) and restart the appliance. When a specific number of fault occurs load get permanently disconnected and locked status is shown via LCD display and LED indicator. All possible condition can be modified by the user using push button and LCD interface.

Result:

Condition	Expected Output	Status
Voltage≤220 && Fault ≤Fix number of fault	Safe load operating green LED on	Achieved
220≥Voltage≤250 && Fault ≤Fix number	Load operating Blue LED on for abnormal	Not Achieved
Voltage≥250 && Fault ≤Fix number of fault	Load Disconnected RED LED on restart when voltage≤250	Achieved
Voltage≤250 && Fault ≥Fix number of fault	Load Disconnected Manual Reset needed	Achieved

Conclusion: The simulated voltage protector project performed close to it expectation. Load disconnection, reconnection, device lockout for safety, user control via hardware device achieved. Limitation in slight voltage serge conditioning and PC control yet not achieved future work will focused on including them.